UTM UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
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Project: Phase 1

Project Proposal & Database Requirement

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1.0 Introduction

Since online shopping has become a trend worldwide, the demand for parcel delivery is getting higher. Nowadays, lots of buyers would choose delivery service for their convenience since delivery saves time and effort. The same situation as in university, students or staff in University Technology Malaysia, Johor Bahru would also choose to use delivery service to pick up parcels from parcel pick up points to their hostel. It is because any of the parcels that enter university would drop off at the respective three parcel points, which are Cengal Parcel Point (CPP), One Stop Parcel Centre (OPC) and ASN Ninja Van Angkasa. These pick up points are three separate locations, which may cause difficulties for those who stay far. Especially someone who does not have personal transportation, they may suffer when picking up their parcels in large sizes, or they are so far away from their destination. The needs of parcel pick up and delivery is significant in such cases to deliver the parcels directly to destination.

Thus, there are some parcel delivery services available in UTM. Initially, they spread the service in public group chats to increase exposure. Then, customers will conduct private conversations with them to notify relevant information. Since the demand keeps increasing, a team has established a website called MALASKETU that allows customers to place orders by filling in google forms. They provide every-day pickup service with reasonable prices listed clearly before customers drop an order.

However, the lack of some services could not satisfy the increasing needs of customers. We want to improve and add new functions by designing a new system so that the business runs well and can grow as time goes by. In this proposal, we will provide our plans for development of the system, by introducing new aspects. For example, tracing of parcels, notification settings and customer service.

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2.0 Background Study

Based on our research and observations, the Malasketu utilises a Linktree page as its homepage, offering four options: parcel delivery service, grab car service, urgent order contact, and parcel delivery proof. When selecting the parcel delivery service option, customers are directed to a Google Form where they input required information. Malasketu uses Google Form as data collection to arrange the parcel pick up and delivery services.

After the customers placed their order, they could not have a look about their parcel status. This will definitely bring inconveniences to the customers like they could not schedule their time to receive their parcel and some might not be available when the driver arrives.

For drivers or someone that is responsible to process the orders, using Google Form is time inefficient. It is because customers are allowed to fill in multiple tracking numbers regardless of pick up points. The drivers have to check for each order and filter out the tracking numbers that have to be picked up at that pick up point.

Other than that, the driver does not get noticed when receiving new orders. This is due to the reason that the customers are placing orders just by filling in a Google Form so the driver will only receive the orders when they manually review the responses.

With the current website, there is no clear classification of orders. When the driver is going to send the parcels, the drivers have to check their destination one by one. This may cause an omission which parcels with the same or nearby destination left out.

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3.0 Problem Statement

1. Unsecured Transaction between Driver and Customer

Currently, monetary transactions between drivers and customers are conducted through some social chat platforms. For example, the driver will send a QR payment code or bank account number to the customer for transferring fees. Payment disputes could arise if transactions are not documented or monitored through an official system. Customers and drivers may have differing interpretations of the terms of the agreement, resulting in misunderstandings and arguments. Not only that, it will cause confusion since it is difficult to hold all parties accountable when disputes arise. Additionally, sending payment details on social chat platforms is vulnerable to fraud, as there are no official records or verification processes to ensure the legitimacy of transactions, making it easier for malicious parties to engage in fraudulent activity.

2. No Specified Mechanism for Reporting Lost and Damage Parcels

When customers encounter a lost or damaged parcel, MALASKETU does not provide a corresponding assistance in coping parcels' issues after delivering. Therefore, they have to contact the driver privately on social chat platforms, which may face risk of encountering drivers' shrinking responsibility. Moreover, it is such an inefficient way for both drivers and customers if a driver has to conduct private conversation with multiple customers. In addition, there is no systematic way for customers to follow up on the status and progress of reported parcel issues in the system. Customers are left without information on the steps taken to resolve the issues they report.

3. Hard to Track Parcel Status

Under the current operating conditions, customers could not get track of their parcel status. For example, they do not know whether their parcel has been picked up by the driver, where the parcel is, when it will be delivered, and so on. It is due to the system's lack of functionality to track the progress of their parcels. They can only know the latest status by frequently asking the delivery person on the chat platform. However, the driver is unable to respond to customer information in a timely manner. This will cause the customers to be confused about the whereabouts of the parcels. Since they have no idea about the status of their packages, it brings uncertainty and inconvenience and also reflects the problem of opaque operations.

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4. Lack of Credibility

Customers do not know the identity of the driver who handles the package, which may cause security issues. For example, some valuable parcels are missing and the corresponding person in charge of the parcels cannot be found for consultation. Anonymity erodes customer trust in the parcel management system. Customers may question the trustworthiness of the individual handling the parcel, which may affect their overall confidence and overall satisfaction in the service. Also, when a driver makes the effort to pick up a parcel as per the customer's request, the customer breaks the appointment such as not paying the fee for parcel delivery, which wastes the driver's time and resources. It can create delays in the entire parcel management process.

5. Lack of Critical Notification

Based on the current operation, customers did not receive any information regarding the transaction unless the parcel had arrived at the destination. There is no notification about the significant process, like who's the driver that is in charge of the delivery, the driver has picked up the parcel, or the driver has set off to deliver. Notification enables efficient communication between driver and customer especially when drivers drop off their parcels. Failure to receive critical notifications may result in communication breakdowns between users and the driver, resulting in misunderstandings or disagreements. The drop off location is a public area which is not secure enough for valuable goods located. Hence, customers would have better arrangements about the pickup time if they could notice the driver's had departed.

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4.0 Proposed Solutions

After discussion among our group members, we decided to develop a comprehensive and innovative system for drivers and customers called ParcelPro. This system is designed to revolutionise the way parcels are managed and delivered within the campus. It provides drivers and customers with an integrated platform that maximises efficiency and improves the transparency of the parcel management process.

Firstly, this system allows both customers and drivers to register and log in by using their user ID and driver ID. The system will require users to fill in their basic information such as phone number, name, and email. Some specific information needs to be filled in by users with different viewpoints. For instance, drivers need to fill in their vehicle type, and vehicle plate as well as upload their photo to enhance their credibility. Similarly, if they are a customer they need to fill in their IC number as a safeguard for both parties. **(login, register, information)-Customers, Drivers**

Moreover, we decided to create an intuitive user interface that allows customers to place an order within the system. This feature will require them to fill in their parcel details such as the tracking no, phone number, receiver name, pickup point, delivery address, the size and quantity of parcel that needs to be picked up. After the customer submits the order, the system will display the order status such as "Order Accepted, Order Rejected, or Pending". If an order has been placed successfully, the users will receive an order number as a reference. **(Order placement)-Customers**

Besides, a two-view instrument panel is provided by the system. The dashboard offers drivers a function of viewing and managing incoming orders. The driver has the option to accept or decline orders based on their ability after viewing the order list. If the order is accepted, the driver can update the parcel status and mark the order as completed, in progress, or pending to delivery. In addition, the system will classify the order list sorted by the hostel and pick-up point to facilitate drivers to access orders. Next, the staff is in charge of monitoring the distribution of incoming orders to drivers. **(Order management)-Drivers, Staff**

Furthermore, we have an idea to design a user-friendly interface within the system where users can easily access their order status. The interface provides a section where users can view the order history and current status of the parcel if the driver accepts the order. For example, it will display "Delivered", "Delivering" and the approximate time of arrival when the driver begins to deliver their parcel. Hence, customers have an idea of the current status of their parcel and would not miss out on picking up it. Not only that, the interface of the order will show the name and photo of the driver, which can improve transparency and credibility. **(Order viewing)-Customers**

This system includes a feature that allows users to report lost or damaged parcels. This can be done by clicking on the respective order in their order history. This portion offers a complaint section for customers. For instance, customers can add pertinent photos and fill in

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the order ID along with information details. Once customers submit their reports, they will receive a reference number for further follow-up. By using the order ID, the relevant person in charge will precisely locate the corresponding order and take the necessary action. At the same time, the subsequent status of the report will be updated in real-time on that page so that users will comprehend the latest status.The staff would assist customers through the complaint-filing process. They will be investigating the issue, communicating with relevant parties, and providing regular updates to customers regarding the status of their complaints to ensure user satisfaction. **(Report)-Customers,Staff**

The system provides notifications to keep users updated on their parcels and services. For example, notifications will be sent to customers when their orders have been received, accepted or not, and when parcels have been picked up or delivered. Users will be notified when parcels have been picked up but payment has not been made, and updates will be provided regularly if users report a parcel lost or damaged, the status of the resolution process. These notifications can be sent via email, SMS, or through the system's user interface. **(Notification)-Customers**

Regarding pickup fees, we have established a systematic payment system, which provides an interface to display a summary of outstanding fees for orders. After the driver collects the parcel, he submits the quantity, size, and weight of the parcel to the system. Then, the system will calculate the total price the customer needs to pay. For the driver, the system will compute the total price after deducting the commission and deposit it into their account, allowing them to withdraw cash at any time. Additionally, customers can pay fees through credit/debit cards and online payment gateways in the system. Online payment provides convenience to students and reduces the time required for the process. **(Fee) - Customers**

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5.0 Objectives

1. To provide a user-friendly and intuitive platform for customers and drivers.

2. To provide a system for reporting lost or damaged parcels and tracking their resolution.

3. To offer real-time order tracking and status updates for users.

4. To provide a transparent transaction process between users.

5. To implement a robust notification system for keeping users informed about their parcels.

6.0 Scope of the Project

The platform that this system aims to use is a mobile app that supports both Android and Apple users. The system will initially prompt users to register and log in to their accounts by entering their ID and password, and then a two-factor authentication mechanism will be activated to verify an identity during a login attempt. When users successfully log in, different corresponding perspectives will be presented to them. Our system is organised into three viewpoints which are for the customer and driver.

Customer's view:

Customers of this system can be anyone in University Technology Malaysia who have the need for parcel delivery. Likely customers are UTM students, employees or family members of an employee. Customers can place an order in the system. If they have already submitted an order, they can only edit the details of the parcel before the order is accepted by a driver. Customers are not allowed to fill in the address outside the UTM area, since parcel delivery services only provided for the area within the UTM campus. Furthermore, customers can view and track the parcel status in real-time from the order. Besides, there is a structured system to allow the customers to make their payments through the system. Only credit card payments, debit card payments, and E-wallet payments are accepted. Not only that, customers are allowed to file a complaint about their parcel issues via the system.

Driver's view:

Drivers of this service are open for anyone who has their own transportation and can meet the driver requirements. Drivers are actually the same group as customers which can be the students, employees or employees' family members who have transportation to provide

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delivery service. Drivers are allowed to have the option to accept or ignore the order. If the driver accepts the order, they may view their incoming orders sorted by the different hostels, which gives the driver an easier way to manage the parcels. Besides, tracking numbers will be sorted according to pick up points. Drivers could have well-arranged information when picking up parcels. Moreover, they can update the current parcel status to customers through the system.

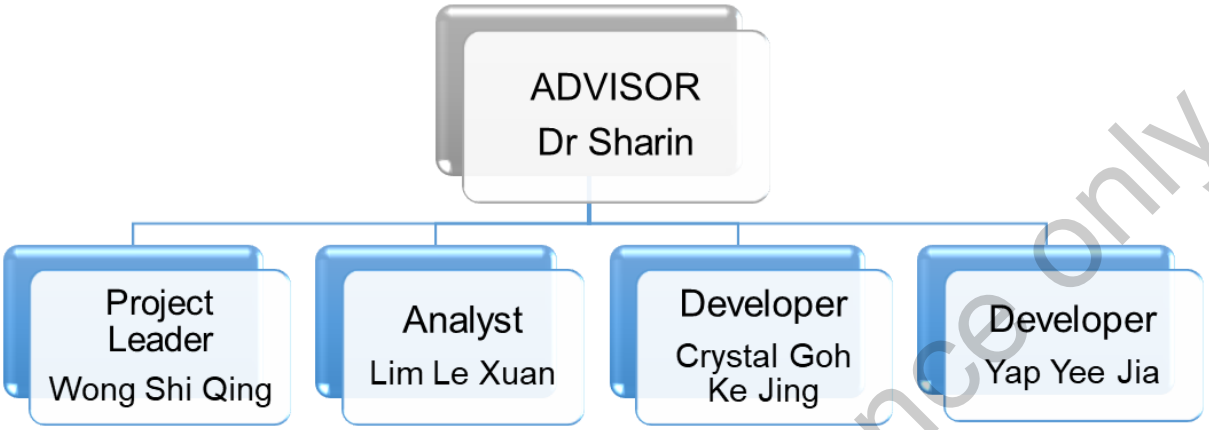
Staff's view:

Driver and customer profiles are viewable, editable, and deactivated by staff. Additionally, it offers tools to verify user information, approve driver registrations, and manage user access levels. Not only that, they offer facilities for managing and keeping track of package orders, such as the ability to allocate orders to drivers and examine all past and current orders. It also has features for monitoring and controlling all financial aspects, such as processing payments, processing bills and invoices, and managing driver compensation. Staff have access to a system that handles customer complaints and disputes in getting their problems resolved.

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7.0 Project Planning

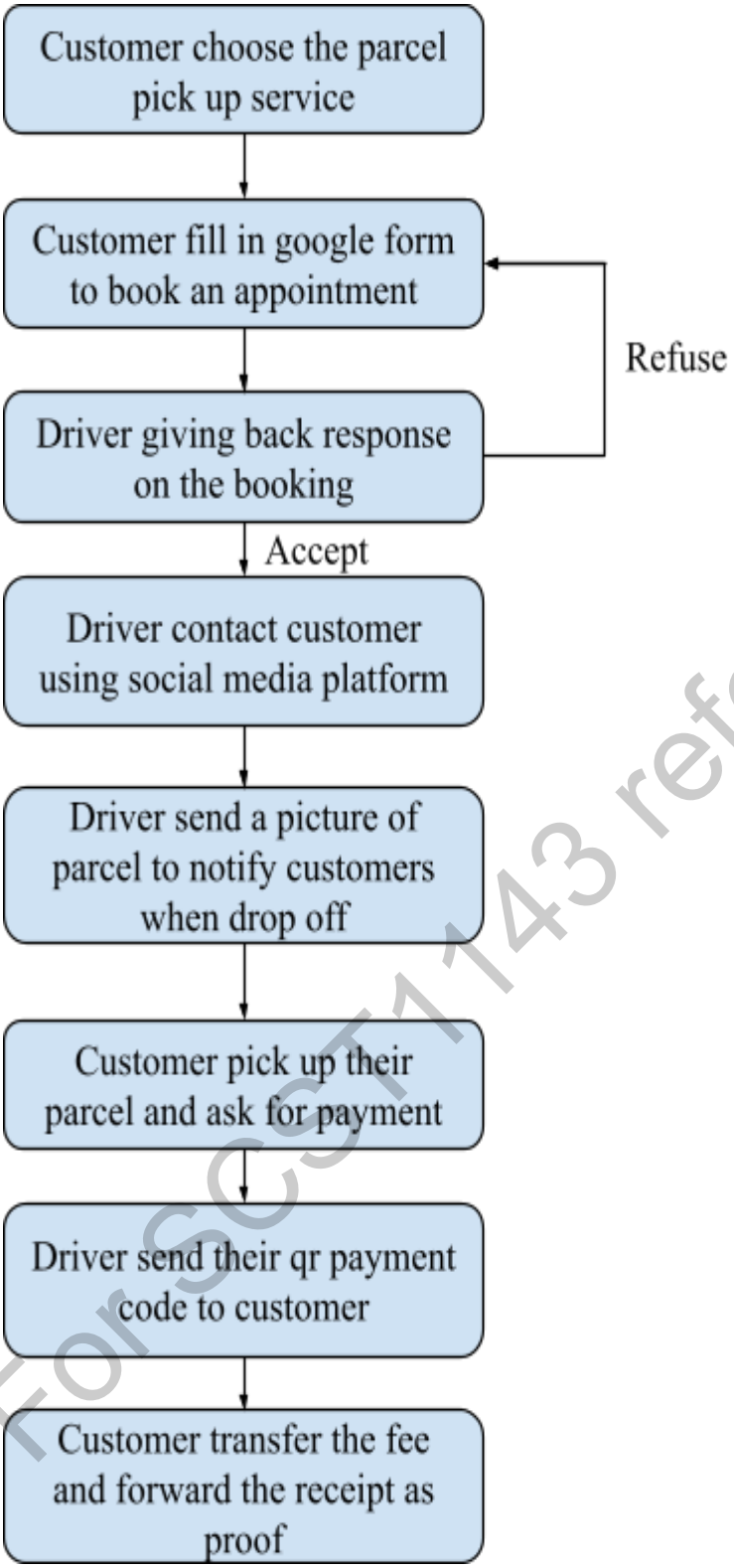
7.1 Human Resource



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8.0 Requirement Analysis

8.1 Current Business Process



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10.0 Benefit and Summary of Proposed System

- Customers can receive notification about parcel status.
- Fees charges will have double verification from both parties.
- Customers can get track of their parcel status including driver's personal details.
- Drivers can pick up parcels with well-arranged order details
- ParcelPro is the system that we proposed to increase the efficiency of process and communication between drivers and customers.
- We decided to propose this system because we observed some problems from the user experience.
- For drivers, they are facing difficulty when picking up parcels for multiple customers at once as the information is not well arranged.
- For customers, they could not get track of their parcel status clearly including the personal details of the driver.
- The proposed solutions aim to resolve these problems by a fast and simple process.
- In order to attract more customers in using this service, we should first improve the user experience.
- Therefore, utilising this system hopes to achieve short and long-term turnover goals by enhancing the user experience.

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